

rename this section to something like model?

0.0.1 Construction of the Hamiltonian

The Hamiltonian is

$$\mathcal{H} = \sum_{\langle ij \rangle} J_{ij} \left(\vec{S}_i \cdot \vec{S}_j + \kappa \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \right) \quad (1)$$

Using the Abrikosov representation of the spin in terms of fermions:

$$\vec{S}_i = \sum_{\alpha\beta} f_{i\alpha}^\dagger \frac{\vec{\sigma}}{2} f_{i\beta} \quad (2)$$

$$\begin{aligned} \Rightarrow \vec{S}_i \cdot \vec{S}_j &= \frac{1}{4} \sum_{\alpha\beta\gamma\delta} f_{i\alpha}^\dagger f_{i\beta} f_{j\gamma}^\dagger f_{j\delta} \underbrace{\vec{\sigma}_{\alpha\beta} \vec{\sigma}_{\gamma\delta}}_{=\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\gamma}\delta_{\beta\delta}} \\ &= \frac{1}{4} \sum_{\alpha\beta} 2f_{i\alpha}^\dagger f_{i\beta} f_{j\beta}^\dagger f_{j\alpha} - f_{i\alpha}^\dagger f_{i\alpha} f_{j\beta}^\dagger f_{j\beta} \end{aligned}$$

Move the last annihilation operator in the first term to the second position: use $f_{j\beta}^\dagger f_{j\alpha} = \delta_{\alpha\beta} - f_{j\alpha} f_{j\beta}^\dagger$ and acquire 1 minus signs. After that bring the other annihilation operator to the last position which gives another minus.

$$\vec{S}_i \cdot \vec{S}_j = \frac{1}{4} \sum_{\alpha\beta} -2f_{i\alpha}^\dagger f_{j\alpha} f_{j\beta}^\dagger f_{j\beta} + 2f_{i\alpha} f_{i\beta} \delta_{\alpha\beta} - f_{i\alpha}^\dagger f_{i\alpha} f_{j\beta}^\dagger f_{j\beta} \quad (3)$$

The second term is n_i and the third is $n_i n_j$. For the first term I introduce the shorthand

$$\chi_{ij} = \frac{1}{2} \sum_{\alpha} f_{i\alpha}^\dagger f_{j\alpha} \quad (4)$$

$$\vec{S}_i \cdot \vec{S}_j = -2\chi_{ij}\chi_{ji} + \frac{1}{2}n_i - \frac{1}{4}n_i n_j \quad (5)$$

Using the constraint $n_i = n_j = 1$, the final expression is

$$\vec{S}_i \cdot \vec{S}_j = -2\chi_{ij}\chi_{ji} + \frac{1}{4} \quad (6)$$

Relevant for the Hamiltonian is also the term:

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j \right)^2 = \frac{1}{16} - \chi_{ij}\chi_{ji} + 4(\chi_{ij}\chi_{ji})^2 \quad (7)$$

The mean field decoupling works like

$$\chi_{ij}\chi_{ji} \approx \langle \chi_{ij} \rangle \chi_{ji} + \langle \chi_{ji} \rangle \chi_{ij} - |\langle \chi_{ij} \rangle|^2 \quad (8)$$

$$(\chi_{ij}\chi_{ji})^2 \approx 2|\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \chi_{ji} + 2|\langle \chi_{ij} \rangle|^2 \langle \chi_{ji} \rangle \chi_{ij} - 3|\langle \chi_{ij} \rangle|^4 \quad (9)$$

$$\Rightarrow \vec{S}_i \cdot \vec{S}_j \approx -2\langle \chi_{ij} \rangle \chi_{ji} + \text{h.c.} + 2|\langle \chi_{ij} \rangle|^2 + \frac{1}{4} \quad (10)$$

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \approx -\langle \chi_{ij} \rangle \chi_{ji} + 8|\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \chi_{ji} + \text{h.c.} + |\langle \chi_{ij} \rangle|^2 - 12|\langle \chi_{ij} \rangle|^4 + \frac{1}{16} \quad (11)$$

Maybe comment of the fact that the mean-field parameters know about the fermi sea and thereby about many states. In this sense the states do speak with each other. Bringing everything together:

$$\begin{aligned} \mathcal{H} \approx \sum_{\langle ij \rangle} J_{ij} & \left[\left(-1 - \kappa/2 + 4\kappa |\langle \chi_{ij} \rangle|^2 \right) \langle \chi_{ij} \rangle \left(f_{j\uparrow}^\dagger f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow} \right) + \text{h.c} \right. \\ & \left. + \left(2 + \kappa - 12\kappa |\langle \chi_{ij} \rangle|^2 \right) |\langle \chi_{ij} \rangle|^2 + \frac{1}{4} + \frac{\kappa}{16} \right] \end{aligned} \quad (12)$$

It is sometimes handy to define

$$\tilde{J}_{ij} = J_{ij} (1 + \kappa/2), \quad \tilde{\kappa} = \kappa / (1 + \kappa/2) \quad (13)$$

I drop the constant:

$$\begin{aligned} \mathcal{H} \approx \sum_{\langle ij \rangle} \tilde{J}_{ij} & \left[- \left(1 - 4\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) \langle \chi_{ij} \rangle \left(f_{j\uparrow}^\dagger f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow} \right) + \text{h.c} \right. \\ & \left. + 2 \left(1 - 6\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) |\langle \chi_{ij} \rangle|^2 \right] \end{aligned} \quad (14)$$

Hence, the mean field produces the effective hopping amplitude t^{MF} :

$$t_{ji}^{\text{MF}} = -\tilde{J}_{ij} \langle \chi_{ij} \rangle \left(1 - 4\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) = -J_{ij} \langle \chi_{ij} \rangle \left(1 + \kappa/2 - 4\kappa |\langle \chi_{ij} \rangle|^2 \right). \quad (15)$$

$$\Rightarrow \mathcal{H} \approx \sum_{\langle ij \rangle} t_{ji}^{\text{MF}} \left(f_{j\uparrow}^\dagger f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow} \right) + \text{const.} \quad (16)$$

This is pretty interesting: the hopping-amplitude $i \rightarrow j$ is directly proportional to the hopping-amplitude $j \rightarrow i$. [Here are a few ideas how to make use/discuss this insight:](#)

- Maybe one can also make the argument that we start with a "symmetric" model, i.e. a model without internal sense of direction, a feature that is now preserved in the Mean field Hamiltonian.
- Or maybe, even more sophisticated, one could make argument using structure of the spin-singlet states.
- Maybe one can use this fact to simplify the Mean-field equation?

0.0.2 Continuity equation and probability current

The probability (/particle) density on site k is

$$n_k = \sum_{\sigma} f_{k\sigma}^\dagger f_{k\sigma}. \quad (17)$$

Now I derive the corresponding probability current density:

$$\frac{\partial n_k}{\partial t} = -i \sum_{\sigma} \left([f_{k\sigma}^\dagger, H] f_{k\sigma} + f_{k\sigma}^\dagger [f_{k\sigma}, H] \right) \quad (18)$$

$$\begin{aligned} [f_{k\sigma}^\dagger, H] f_{k\sigma} &= - \sum_{\delta} t_{k+\delta; k} f_{k+\delta; \sigma}^\dagger f_{k\sigma} - \mu_k f_{k\sigma}^\dagger f_{k\sigma} \\ f_{k\sigma}^\dagger [f_{k\sigma}, H] &= \sum_{\delta} t_{k; k+\delta} f_{k\sigma}^\dagger f_{k+\delta; \sigma} + \mu_k f_{k\sigma}^\dagger f_{k\sigma} \end{aligned} \quad (19)$$

$$\Rightarrow \frac{\partial n_k}{\partial t} = - \sum_{\delta} i \left(\sum_{\sigma} t_{k,k+\delta} f_{k\sigma}^{\dagger} f_{k+\delta;\sigma} - t_{k+\delta,k} f_{k+\delta;\sigma}^{\dagger} f_{k\sigma} \right) \quad (20)$$

Define

$$J_{k \rightarrow k+\delta} \equiv i \left(\sum_{\sigma} t_{k,k+\delta} f_{k\sigma}^{\dagger} f_{k+\delta;\sigma} - t_{k+\delta,k} f_{k+\delta;\sigma}^{\dagger} f_{k\sigma} \right) \quad (21)$$

$J_{k \rightarrow k+\delta}$ is the (directed!) current density along the bond $\vec{R}_k \rightarrow \vec{R}_{k+\delta}$. The discrete (/lattice) continuity equation reads

$$\frac{\partial n_k}{\partial t} = - \sum_{\delta} J_{k \rightarrow k+\delta}. \quad (22)$$

Something interesting happens when calculating $\langle J_{k \rightarrow k+\delta} \rangle$:

$$\begin{aligned} \langle J_{k \rightarrow k+\delta} \rangle &= i \left(t_{k,k+\delta} \left\langle \sum_{\sigma} f_{k\sigma}^{\dagger} f_{k+\delta;\sigma} \right\rangle - t_{k+\delta,k} \left\langle \sum_{\sigma} f_{k+\delta;\sigma}^{\dagger} f_{k\sigma} \right\rangle \right) \\ &= 2i (t_{k,k+\delta} \langle \chi_{k,k+\delta} \rangle - t_{k+\delta,k} \langle \chi_{k+\delta,k} \rangle) \\ &= -2i \tilde{J}_{k;k+\delta} \left(1 - 4\tilde{\kappa} |\langle \chi_{k,k+\delta} \rangle|^2 \right) \left(\langle \chi_{k+\delta;k} \rangle \langle \chi_{k;k+\delta} \rangle - \langle \chi_{k;k+\delta} \rangle \langle \chi_{k+\delta;k} \rangle \right) \\ &= 0. \end{aligned} \quad (23)$$

Apparently, the current on each bond vanishes in the ground state by construction. **This still leaves the possibility that single states have non-zero probability currents. However, it remains to investigate how this is possible for e.g. states with a singular (topologically protected) edge mode. One would guess that this feature would lead to a non-zero current in the ground state.**